

Acoustic communication in the Asian elephant, *Elephas maximus maximus*

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Summary

Existing knowledge of acoustic communication in elephants is based primarily on African species (*Loxodonta africana* and *Loxodonta cyclotis*). There has been comparatively less study of communication in Asian elephants (*Elephas maximus*). In order to provide a basis for understanding the evolution and function of acoustic communication in proboscideans, I present a quantitative description of vocal communication in wild Asian elephants. I classify calls by acoustic features into 8 'single' calls, 5 'combination' calls and one possibly unique male call for a total of at least 14 distinct call types. Some of these vocalizations have never before been described. Certain low-frequency calls are individually distinct. Acoustic signals occur in a wide range of social contexts, with some differences in call production among age and sex classes.

Keywords: Asian elephant, *Elephas maximus*, acoustic signals, vocal communication, vocal repertoire.

Introduction

Similarities in the communication systems of distantly related taxa may be due to similarities in socio-ecology. Development of general frameworks to explain convergence and divergence in the evolution of communication systems first requires the ability to quantitatively compare across the signaling systems of disparate taxa. Acoustic signals are used in searching and competing for mates (e.g., crickets: Mhatre & Balakrishnan, 2006; frogs: Ryan & Rand, 2003; birds: Mountjoy & Lemon, 1996). Distinct vocalizations are

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used by both bats (Brown et al., 1983; Bohn et al., 2007) and penguins (Aubin et al., 2000) to locate specific individuals in crowded colonies. In these contexts, acoustic signals aid navigation and spacing of individuals.

Acoustic signals also serve social functions. Social species typically have some mutual interest in being together (e.g., for defense, hunting, holding territory, or rearing offspring) and interact repeatedly. Hence, social input shapes communication among primates (McComb & Semple, 2005), cetaceans (Connor et al., 1998) and birds (Nordby et al., 2000). Vocalizations help maintain social bonds and indicate rank in baboons (Cheney et al., 1995; Kitchen et al., 2003) as well as spotted hyenas (Holekamp et al., 1999, 2007), both of which have societies with linear dominance hierarchies.

Asian elephants (*Elephas maximus*) belong to the Proboscidean clade, whose only other extant members are the African savannah elephant (*Loxodonta africana*) and the African forest elephant (*Loxodonta cyclotis* or *Loxodonta africana cyclotis*) (Shoshani & Tassy, 1996). Although the species status of African forest elephants is much debated (Roca et al., 1999, 2007; Eggert et al., 2002; Debruyne, 2005; Johnson et al., 2007), the Asian and African elephant species are separated by at least six million years (Shoshani & Tassy, 1996; Rohland et al., 2007) and social organization may differ among them. Savannah elephant family units led by the eldest adult females (matriarchs) form bonds with other families throughout their home ranges. Wet season aggregations break up during the dry season migrations, giving rise to a fission–fusion society with hierarchical ‘tiers’ (Douglas-Hamilton, 1972; Moss & Poole, 1983; Wittemyer et al., 2005). This suggests that companionship is preferred by African savannah elephants, as long as resources permit it. Moreover, matriarchs serve an important function by leading their groups to scarce resources by memory (McComb et al., 2001). There is no evidence of such migration by Asian elephants (Fernando et al., 2008). Past studies of Asian elephants also indicate low rates of association among relatives (Fernando & Lande, 2000; Vidya & Sukumar, 2005). It has been suggested Asian elephants do not appear to associate beyond the family level (McKay, 1973; Fernando & Lande, 2000; Vidya & Sukumar, 2005). If Asian elephants do not have a high affinity for conspecifics, one might expect fewer or less diverse acoustic signals serving social functions.

African savannah elephants use acoustic signals for mate-search (Poole et al., 1988), male–male competition (Poole, 1999), and maintenance of social bonds (Poole et al., 1988). They detect low-frequency calls over a range

of several kilometers (Langbauer, 1991, 2000; Garstang, 2004). Such calls also mediate inter- and intra-group social encounters (Berg, 1983; Poole et al., 1988; McComb et al., 2000, 2003; Leong et al., 2003a,b; Soltis et al., 2005a,b; Wood et al., 2005; O'Connell-Rodwell et al., 2006; Leighty et al., 2008). Less is known about the function of vocalizations in African forest elephants, though calls are used to census remote populations (Payne et al., 2003). A partial vocal repertoire for Asian elephants has been verbally described (McKay, 1973), and some acoustic (Payne et al., 1986; Nair et al., 2009), as well as seismic (O'Connell-Rodwell et al., 2000) features of vocalizations are documented. This study provides a quantitative description of acoustic signals produced by the Sri Lankan elephant (*E. maximus maximus*), including previously undescribed vocalizations.

Methods

Study site and subjects

Uda Walawe National Park (UWNP), Sri Lanka, is located at latitude 6°30'14.0646"N, longitude 80°54'28.1268"E, and an average altitude of 118 m above sea level. It occupies 308 km² and contains tall grassland, dense scrub, riparian forest, secondary forest, rivers and seasonal streams. It also contains several natural and man-made water sources and reservoirs with seasonal floodplains. There are two monsoons per calendar year, separated by dry seasons of variable length. Over 300 adult females have been individually identified in UWNP using characteristics of the ears, tail, and other natural markings (Moss, 1996).

Data collection

Data were collected from May, 2006 to December, 2007. Observations were performed by vehicle during park hours from 0600 to 1830 h. We photographed or videotaped all adult females upon encounter. A 'group' was defined as animals within visual range of the observer (up to 500 m) that were either moving or resting together. Identities of known individuals were noted in addition to the number of adult females (>10 years and has been pregnant), sub-adults (8–12 years), juveniles (3–7 years), infants (6 months–2 years) and newborns (<6 months). These size classes were based on height

relative to that of an adult female. Corresponding ages were based on personal observations of wild calves of known age as well as animals of known age reared in captivity.

Recordings of vocalizations were made using an Earthworks QTC50 microphone shock-mounted inside a Rycote Zeppelin windshield, via a Fostex FR-2 field recorder (sampling rate 48 kHz) connected to a 12 V lead acid battery. Recordings were initiated at the start of a call with a 10-s pre-record buffer so that the entire call was captured and loss of rare vocalizations minimized. This was made possible with the 'pre-record' feature of the Fostex, which records continuously, but only saves the file with a 10-second lead once the 'record' button is depressed. In order to minimize loss of low-frequency or potentially inaudible calls, recording was continued for at least three minutes following the end of vocalization events. During the first two months, hour-long recording sessions were also carried out opportunistically while in close proximity to a group. However, spectrograms showed that few vocalizations were captured; therefore, this was discontinued. Extensive vocal activity and rare occurrences were also video recorded. When possible, distance from microphone to vocalizer was noted. Distances of 20 m or less were visually estimated by the observer, greater distances were measured using a laser range finder, with accuracy of ± 1 m.

Behavioural data were collected with an HP iPaq 1945 hand-held pocket computer and custom software. Ad libitum observations during vocal events were whispered into a dictation microphone-mask, which fed into a second audio channel. This enabled precise time-synchronization between recordings and behavioural observations, as well as clearer tagging of vocalization to callers where caller ID was unambiguously observed. Keeping human speech on a separate channel from target recordings prevented interference with audio recordings. If there were multiple callers, callers were identified if they showed obvious behavioural cues such as an open mouth, lifted tail and head, or flapping ears. If there was ambiguity between callers, the probability of an individual calling was noted. For instance, if a call could have been produced by one of two individuals, the ID probability was listed as 1/2 and if it could have been produced by one of three, it was 1/3, etc. No overlapping calls were used in further acoustical analyses. Only calls where the ID certainty was 1 were used in analysis of individual variation.

Thirty-minute focal animal samples of known animals were initiated upon positive identification (Altmann, 1974). Vocalizations produced by non-focal

animals were audio-recorded and noted *ad libitum*. In sequentially sampling multiple individuals, it was possible to spend several hours with a single group and, thus, observe rare vocalizations, long call bouts, or periods of collective vocal activity. The ethogram consisted of feeding behaviours (grazing, browsing), orientation responses, listening, smelling (trunk held in an 'S' or 'J' position or toward some object), aggression (with or without body contact), movement (general movement, passing or approaching a specific individual, leaving, brisk walk, running), social interactions (touching the trunk to face or body parts), dominance or threat displays (trunk over another individual, tossing soil or vegetation over the back during a disturbance, charging, pushes, bites), mating (mounting), nursing, playing, urinating, defecating and vocalizations. Start and stop times were recorded for resting, water and mud activity, and defensive huddling.

Contexts of vocalizations were broadly defined using 16 months of behavioural data from September, 2006 to December, 2007. Situations in which one or more individuals showed directed movement toward or away from one another, orientation or listening responses, or non-aggressive bodily touches were subsumed under 'inter- or intra-group social interaction'. The context of 'searching' was characterized by the subject exhibiting orientation, listening and trunk lifting. 'Movement' was simply physical movement. 'Fear' was characterized by the subject showing widened eyes, brisk movements away from a source of disturbance or defensive huddling. 'Excitement' was characterized by any of the following: urination and defecation, ear flapping, widened eyes, temporal secretion, brisk movement with head, ears or tail held high, spinning, head tossing. 'Aggression' involved exhibition of aggression, threat, or dominance behaviours. 'Disturbed' encompassed all other situations in which vocalizations followed external disturbance (predators, startled animals, humans), or in which the subject appeared agitated. 'Other' comprised situations that rarely involved vocalizations, such as play, nursing, or mating. These contexts are not mutually exclusive. For instance, a call might be produced while moving with or without search behaviour.

Audio data annotation

Spectrograms of calls were first annotated manually on Praat v4.5.16 (Boersma & Weenink, 2009) and later automatically extracted according to labelled segment boundaries.

Acoustic features

Periodic and a-periodic calls

Acoustic analyses were not performed on vocalizations from males as these were rare. Only calls of exemplary acoustic quality were used, as judged by the visibility of all measured components of the call on a spectrogram and the accuracy of automated measurements. This excludes overlapping calls or those with extrinsic noise.

‘Harmonicity’ quantifies the periodic nature of calls, expressed in dB (Boersma, 1993; Boersma & Weenink, 2009). A lower harmonicity indicates less periodicity or lower signal-to-noise ratio. Because this ratio changes over the course of a single call, average and standard deviation of harmonicity over an entire call were calculated from intervals of 0.05 s for all calls except squeaks, for which I used intervals of 0.005 s as they were otherwise too brief.

For all other measurements, vocalizations that showed clear fundamental frequencies (periodic) were analyzed separately from those that did not (a-periodic). Calls that did not have stable periodic structures were classified by duration alone. Vocalizations by juveniles were included as some calls types were primarily produced by calves.

For periodic calls, measurements were taken with Praat, using the program’s built-in pitch tracker which identifies the fundamental (F_0) frequency. Calls from juveniles were excluded as pitch is likely to change with age. Settings for accurate pitch extraction were calibrated separately for each call category by visually inspecting the accuracy of the pitch line generated by Praat on spectrograms. Once pitch detection settings giving 100% accuracy were determined, eight measurements on the fundamental frequency contour for each call were automated: duration, minimum, time at minimum, maximum, time at maximum, mean, standard deviation and absolute slope (Boersma & Weenink, 2009). From these measures, I further calculated percent at maximum (elapsed time at maximum/duration), percent at minimum (elapsed time at minimum/duration) and range (max–min).

Combination calls

Some vocalizations showed periodic and a-periodic regions as at least two distinct segments. These segments resembled calls that also occurred by themselves. I measured each segment separately and compared these to the

calls that occurred singly to determine whether they could accurately be described as ‘combinations’ of calls. The first segment was almost always a-periodic, hence only duration and harmonicity were measured. The second segment was distinctly periodic, hence all features were measured.

Statistical analysis

Statistical analyses were performed on Matlab v.7.0. I used the Mann-Whitney *U*-test to compare harmonicities and durations of single call types with the corresponding segments of combination calls. I also differentiated two classes of low-frequency calls by eye first on the presence or absence of harmonics above 500 Hz, and then compared their standardized F_0 features with the Mann-Whitney *U*-test. Two-tailed *t*-tests were used for comparing acoustic features of calls with sufficiently large and evenly matched sample sizes. Where there were multiple calls of a certain type from the same individual, only the mean value of an acoustic feature was used for that individual in order to avoid pseudoreplication. For the most common type of vocalization, I used MANOVA to test whether calls from certain identified individuals were distinctive based on the same F_0 measures. I further examined the acoustic similarity of low-frequency calls and call segments using principal components analysis.

Results

In total 3921 calls were recorded and annotated from the period of January–December, 2007. They were classified into 14 categories, out of which growls were the most commonly recorded, comprising almost two thirds. Table 1 summarizes the distribution of vocalization types, abbreviations used throughout the paper, and general contexts. In all tables, N_{calls} is the number of measurable vocalizations whereas N_{ind} is the number of individual vocalizers responsible. Not all calls could be used for all measurements.

Harmonicity

The medians, ranges, and sample sizes of calls with measurable harmonicity are reported in Table 2. Growls (using the term by McKay, 1973) were the most harmonic. Harmonicities of the first component of combination

Table 1. Summary of calls and contexts.

Call	Abbr.	<i>N</i> _{tot}	% of total	Str.	<i>N</i> _{ax}	Context (% calls)													Age and sex class (% vocalizers)							
						1	2	3	4	5	6	7	8	9	10	11	12	13	AF	AM	SF	SM	SB	JV	IN	
Growl	GRW	2580	65.8	S, R, C	187	–	11.8	–	5.9	7.5	26.7	53.2	38.8	8.6	–	–	1.3	–	79.1	1.1	8.0	2.1	3.7	5.3	<1	
	SQK	343	8.75	R	67	1.5	20.9	–	14.9	31.3	29.9	17.9	10.4	3.0	–	–	6.0	1.5	32.8	–	19.4	19.4	7.5	17.9	3.0	
	LRM	225	5.74	R, C	31	3.2	–	–	9.7	–	19.4	9.7	54.8	9.7	9.7	–	–	9.7	41.9	3.2	–	3.2	9.7	38.7	3.2	
Longroar-rumble																										
Longroar	LRR	158	4.03	S, R, C	45	–	–	2.2	–	11.1	35.6	–	46.7	42.2	2.2	–	6.7	6.7	11.1	–	20.0	–	–	–	68.9	–
Rumble	RUM	151	3.85	S, R, C	103	–	6.8	–	3.9	5.8	34.0	40.8	35.9	8.7	–	–	2.9	<1	72.8	1.9	11.7	–	2.9	9.7	<1	
Bark-rumble	BRM	133	3.39	R, C	43	–	2.3	–	11.6	–	18.6	11.6	74.4	30.2	–	–	2.3	2.3	53.5	2.3	7.0	–	4.7	32.6	–	
Trumpet	TMP	129	3.29	S	74	1.4	48.6	–	21.6	10.8	10.8	7.4	8.1	6.8	–	–	4.1	5.4	33.8	4.1	21.6	5.4	4.1	23.0	8.1	
Roar-rumble	RRM	68	1.73	R, C	16	–	–	–	–	–	18.8	18.8	37.5	12.5	–	–	6.3	18.8	43.8	6.3	18.8	–	3.1	28.1	–	
Roar	ROR	45	1.5	S	66	–	4.5	1.5	6.1	7.6	25.0	–	40.9	16.7	1.5	–	8.3	1.5	28.8	4.5	10.6	9.1	–	40.9	6.1	
Bark	BRK	39	0.99	S	15	33.3	13.3	–	–	–	20.0	–	13.3	–	–	–	6.7	20.0	40.0	20.0	6.7	–	6.7	26.7	–	
Squeal	SQK	33	0.84	S, R	6	–	–	–	66.7	33.3	–	–	–	–	–	–	–	–	66.7	–	33.3	–	–	–	–	
Croak-rumble	CRM1	15	0.42	R, C	11	–	27.3	–	18.2	27.3	–	63.6	36.4	9.1	–	–	–	–	56.4	–	18.2	1.8	–	–	–	
Chirp-rumble	CRM2	2	<0.4	R	3	–	–	–	–	–	–	66.7	–	–	–	–	–	–	33.3	100	–	–	–	–	–	
Musth chirp-rumble	MCR	3	<0.4	R	4	–	–	–	–	–	–	–	–	–	–	100	–	–	–	100	–	–	–	–	–	

N_{tot} is the total number of recordings of each type, whereas N_{ex} is the number of calls assessed for context. Only one call type per age class was counted during any single event in determining contexts. However, contexts are not mutually exclusive hence percentages do not sum to 100. The most common contexts for particular call types are in boldface. Str is the temporal structure of calls, where S means that over 90% of the time, call occurs singly; R means that over half the time calls are repetitive and may occur in a bout (where the interval between calls is shorter than the duration of a single call), and also that multiple bouts may occur successively (where the interval between bouts is greater than the interval between calls within a bout); and C means that multiple individuals may call simultaneously, hence chorusing. Contexts: 1, vocalizer exhibits aggression with physical contact; 2, vocalizer exhibits aggression without contact (threats); 3, vocalizer receives aggression with or without physical contact; 4, fear; 5, excitement; 6, disturbance; 7, non-aggressive social; 8, movement; 9, searching; 10, being dominated or coerced by another; 11, musth; 12, other: nursing, play, mating; 13, unknown. Age and sex classes are adult female, adult male, sub-adult female, sub-adult male, sub-adult of undetermined sex, juvenile and infant.

Table 2. Harmonicities of calls and call segments.

Call	N_{calls}	N_{ind}	Median of average	Interquartile range of average	min_{avg}	max_{avg}	Median of stDev	Interquartile range of stDev	$\text{min}_{\text{stdev}}$	$\text{max}_{\text{stDev}}$
SQK	108	15	13.67	13.06	4.33	30.55	13.06	5.46	4.33	30.55
SQL	20	7	12.18	6.67	7.33	17.76	5.30	6.14	2.61	16.73
TMP	26	13	8.92	5.46	5.52	19.14	7.57	5.46	1.98	19.14
RUM	22	11	11.40	5.15	9.26	16.13	5.15	5.46	3.66	16.01
GRW	33	11	11.89	6.03	8.61	24.30	6.03	5.46	3.10	24.30
BRM1	12	7	5.18	5.18	3.78	12.17	5.18	6.01	2.93	7.29
BRM2	8	5	8.49	8.49	5.16	24.24	8.49	2.50	4.33	24.24
BRK	5	5	8.93	8.93	2.91	11.60	8.93	3.86	2.91	11.60
LRM1	9	7	8.38	7.58	2.06	15.15	7.58	3.86	2.06	15.15
LRM2	8	6	6.18	3.67	1.78	12.57	3.67	3.02	2.42	12.57
LRR	33	22	11.85	7.62	1.95	29.04	7.62	3.02	1.95	29.04
ROR	14	13	12.48	12.48	-1.92	20.85	12.48	3.10	-1.92	20.85
RRM1	3	3	-0.38	-0.38	-0.80	10.21	4.15	3.26	-0.80	10.21
RRM2	3	3	9.29	9.29	8.68	16.43	9.29	3.26	8.68	16.43

Compound calls such as BRM are broken into segments such as BRM1 and BRM2.

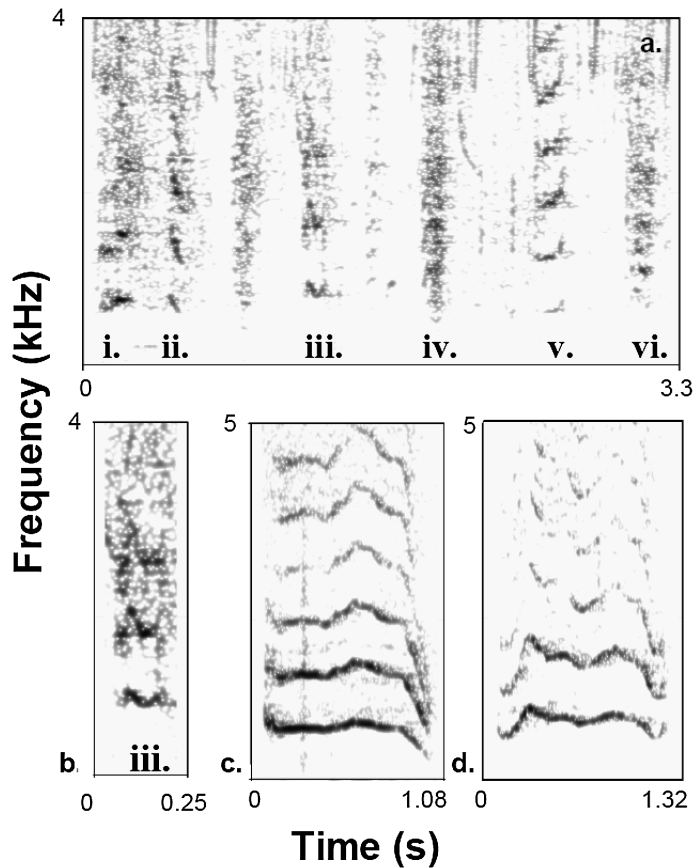


Figure 1. High-frequency calls. (a) Bout of squeaks (SQK) from same individual shows varied contours; (b) squeak (iii) exemplifies the most typical u-shaped frequency contour; (c, d) two squeals (SQL) from one individual during the same recording session.

calls were not significantly different from those of corresponding single calls, nor did growls differ from rumbles (two-tailed Mann-Whitney *U*-test, $p > 0.05$). Squeaks (Figure 1a,b), squeals (Figure 1c,d), trumpets (Figure 2d), growls (Figures 3a,b and 4a,b) and rumbles (Figures 3c,d and 4c,d) all showed higher harmonicity (> 10 dB) than barks (Figure 2a) or roars (Figure 2c). Longroars (Figure 2b), though very noisy vocalizations, did not differ significantly in harmonicity from growls (two-tailed Mann-Whitney *U*-test, $p = 0.98$) (Figure 5). Also in contrast to expectation, trumpets showed low harmonicities overall. Harmonicity captures periodicity that is not solely a feature of the sound source (fundamental frequency) but also of the filter

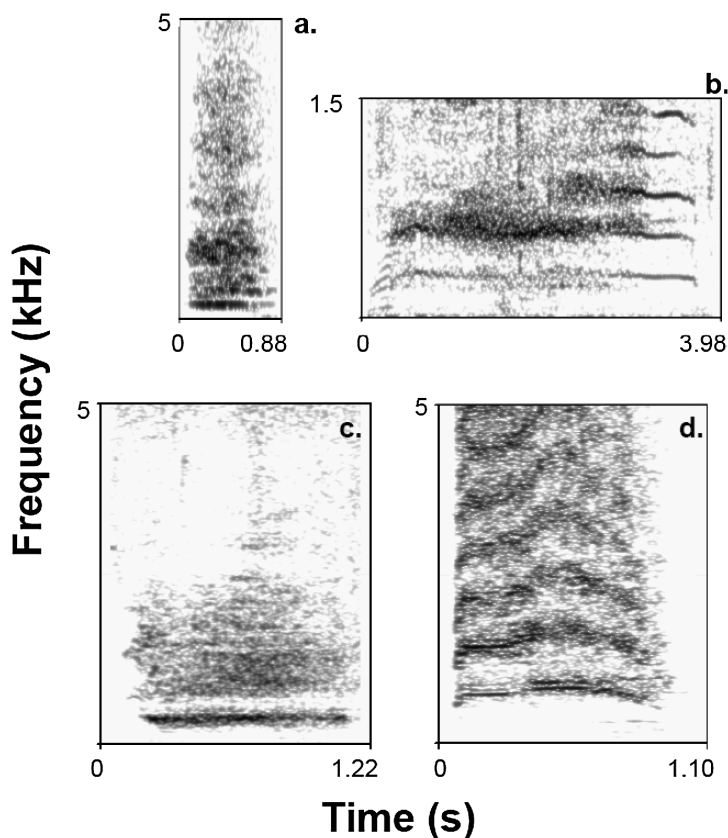


Figure 2. Calls with a-periodicities. (a) Bark (BRK); (b) long roar (LRR) showing growl-like onset, chaotic region and high-frequency harmonics; (c) roar (ROR); (d) trumpet with double-voicing.

(resonances). Longroars, thus, may not have a clear fundamental frequency, but do show broad resonant regions, and calls of some individuals contain partial harmonic regions (Figure 2b).

A-periodic calls and segments

A-periodic calls were graded by duration (Figures 2 and 8), with ‘Barks’ being shortest and ‘Longroars’ being longest. The duration of the first component of combination calls (Figures 6–8) were not significantly different from that of single calls for bark-rumbles (two-tailed T -test, $N_{\text{BRM}} = N_{\text{BRK}} = 24$, $p = 0.275$) or roar-rumbles ($N_{\text{RRM}} = N_{\text{ROR}} = 28$, $p = 0.437$), but was sig-

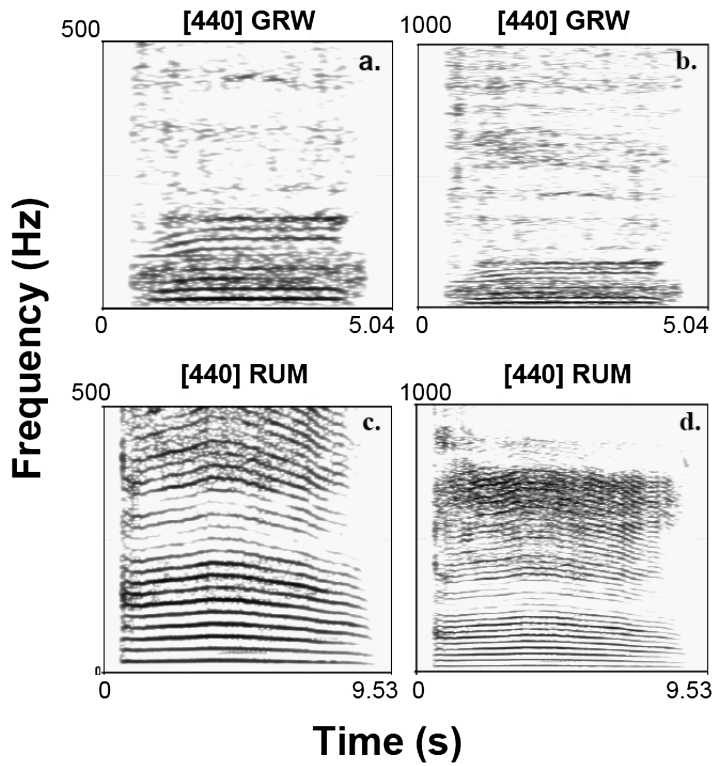


Figure 3. Growls vs. rumbles by individual [440]. (a, b) Single growl (GRW) by adult female [440] at 500 and 1000 Hz, recorded at 3 m; (c, d) rumble (RUM) by the same individual, same recording session, at 500 and 1000 Hz, recorded at 10–15 m distance.

nificant for longroar–rumbles ($N_{\text{LRM}} = N_{\text{LRR}} = 30$, $p < 0.05$). ‘chirps’ and ‘croaks’ were <1-s-long elements that were more tonal than barks and were always joined to a rumble or growl-like segment. The resulting compound calls were termed ‘chirp–rumbles’ and ‘croak–rumbles’ (Figure 7). Chirps and croaks together are included for comparison with durations of all calls in Figure 8.

Periodic calls and segments

Table 3 summarizes all pitch contour measurements on calls and call segments that showed clear F_0 frequencies. ‘Growls’ were defined by eye as calls having little or no energy above 500 Hz, as opposed to ‘rumbles’, which contain energy up to 1 kHz. This was not due to the attenuation of higher frequencies at greater recording distances, as both types have been recorded at

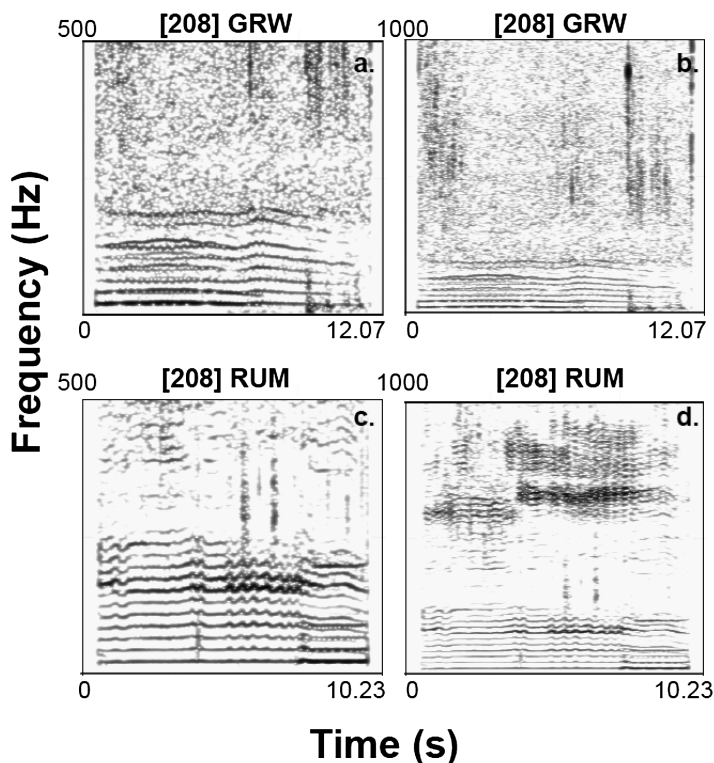


Figure 4. Growls vs. rumbles by individual [208]. (a, b) Single growl by adult female [208] at 500 and 1000 Hz, recorded at 10 m, slight overlapping call from another individual; (c, d) rumble by the same individual, same recording session, at 500 and 1000 Hz, recorded at 10–15 m distance.

close range from identified individuals (Figures 3 and 4). Moreover, rumbles could be distinguished from growls based on some but not all F_0 measures (Figure 5). Growls were individually distinct (Figure 9).

The second segment of combination calls were also initially termed ‘rumbles’ because many showed frequencies above 500 Hz and resembled rumbles to the human ear. However, principal components of fundamental frequency and duration measures distinguished these ‘rumbles’ of compound calls from either growls or rumbles which occurred independently (Figure 10, Table 4). The visibility (and audibility) of upper harmonics may be dependent on recording distance, however, samples with known identities and distances were too few to test this relationship.

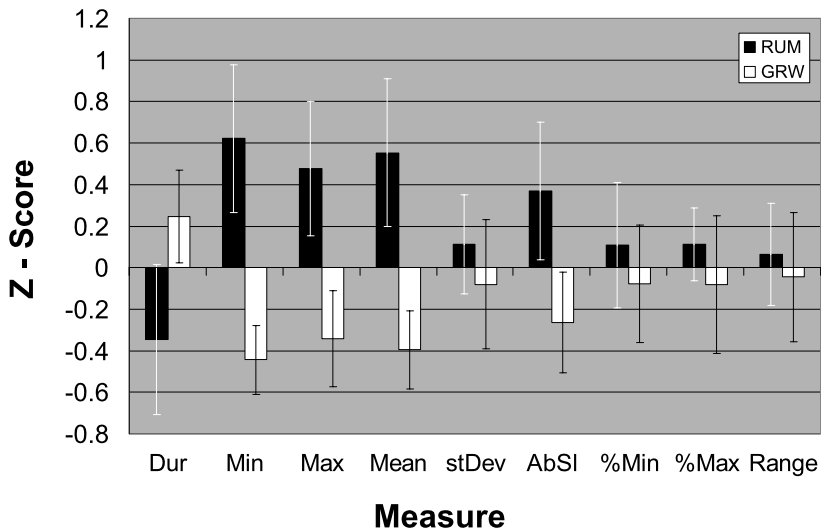


Figure 5. Growls and rumbles are distinguishable by F_0 features. $N_{RUM} = 21$ calls, 10 individuals, $N_{GRW} = 39$ calls, 14 individuals. Plot shows standardized F_0 measure averages $\pm$ standard error for each. The two call types differed most in their minima, maxima, means and absolute slope (two-tailed Mann-Whitney: $U_{min} = 25$, $p_{min} = 0.009$, $U_{max} = 37$, $p_{max} = 0.057$, $U_{Mean} = 33.5$, $p_{Mean} = 0.035$, $U_{AbSlope} = 33.5$, $p_{AbSlope} = 0.046$). Growls are lower frequency vocalizations than rumbles, with a tendency towards longer duration. Rumbles show more absolute changes in frequency from start to finish (rise and fall).

Context

Rumbles and growls

Observations of calls by context are summarized in Table 1. Growls were given in almost all social situations. They typically occurred in bouts or choruses by several individuals while oriented towards an object of attention. Visual cues of growls were depressed cheeks while the mouth remained almost closed while ears were usually outspread and still. Growls were not audible to us beyond 20 m, especially under windy conditions. On at least four different occasions they were produced repeatedly by adult females who appeared to be grazing or moving alone. On another occasion an adult female uttered a growl as she began to walk away from a sub-adult female investigating our jeep. The adult paused with her back towards us as they briefly exchanged growls that were barely audible, after which the younger individual turned and followed. Only once was an exchange of growls observed among adult males.

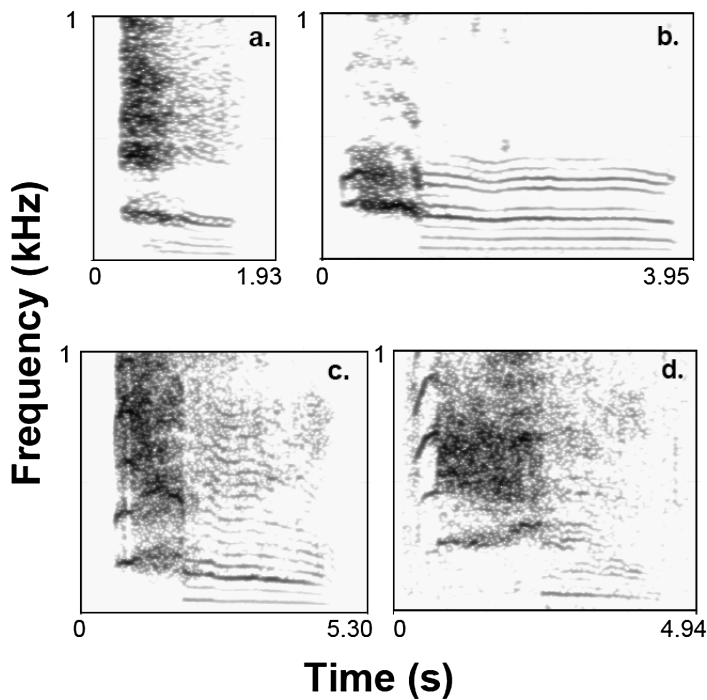


Figure 6. Combination calls with a-periodicities in leading segment. (a) Bark-rumble (BRM); (b) roar-rumble (RRM); (c) long-roar-rumble (LRM) in which first segment is shorter than the second; (d) long-roar-rumble in which first segment is longer than the second.

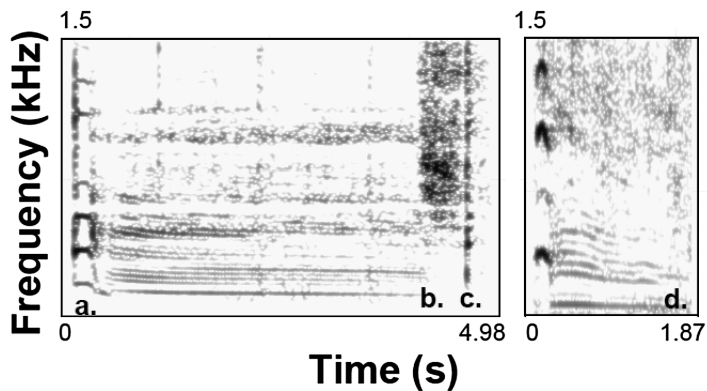


Figure 7. Combination calls with periodic leading segments and non-vocal acoustic signals. (a) Croak-rumble; (b) blow; (c) trunk-bounce; (d) chirp-rumble.

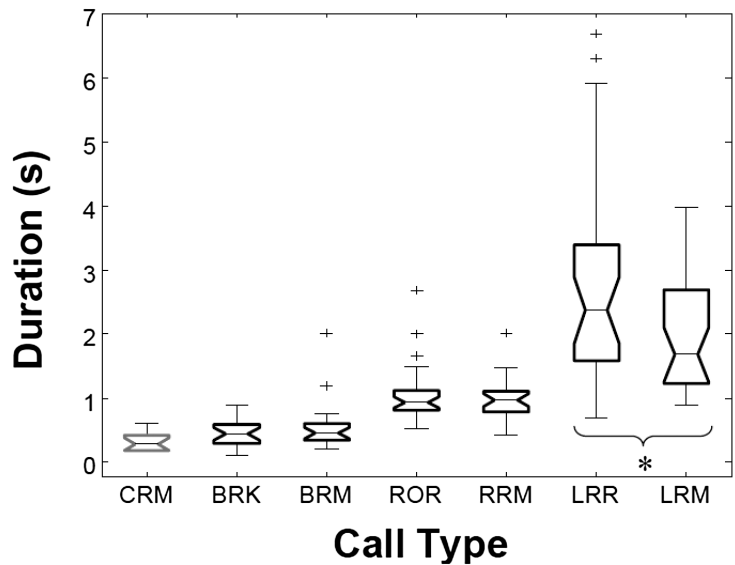


Figure 8. Durations of the first segment of combination calls and corresponding single calls. Durations of the leading segment of chirp- and croak-rumbles (CRM), bark-rumbles (BRM), roar-rumbles (RRM) and longroar-rumbles (LRM) compared to the duration of barks (BRK), roars (ROR) and longroars (LRR). Asterisk indicates significant difference (see text).

Rumbles (using the term of Poole et al., 1988) occurred in contexts similar to growls. They most frequently occurred in choruses, with calls by two or more individuals creating a continuous sound. Single rumbles in such cases were difficult to isolate. Rumbles seemed louder than growls to human ears, and were accompanied by more obvious visual cues such as slightly lifted head, rapidly flapping ears and open mouth. They were audible at distances of at least 500 m, even with wind. No calls resembling rumbles or growls were recorded from infants under 2 years of age. Adult female vocalizers were typically approached by others. In one instance, an adult female gave a series of rumbles and was approached by a second adult female from 100 m away. This female gave a trunk-bounce, touched her several times, and then left again, but was not followed by the initial caller. Rumbles or growls by juveniles or sub-adults did not elicit approach from adults, though they did sometimes elicit approach and exploratory behaviour from young. On two occasions growls and rumbles were observed to maintain distance between un-affiliated social groups. Individuals also sometimes moved towards growls and rumbles whose source was obscured by vegetation.

Table 3. F_0 features of calls showing harmonic structure.

Call	N_{calls}	N_{ind}	Duration	min	max	Mean	stDev	Abs slope	% min	% max	Range (max-min)
SQK	107	15	0.24 ± 0.01	1079.39 ± 40.85	1208.84 ± 48.36	1133.42 ± 43.04	41.17 ± 5.51	742.51 ± 76.26	44.66 ± 2.99	54.97 ± 3.91	129.45 ± 17.48
SQL	15	7	1.08 ± 0.09	646.85 ± 56.03	1041.80 ± 79.59	906.33 ± 70.24	103.24 ± 15.20	680.42 ± 153.35	79.12 ± 7.51	29.86 ± 5.40	394.95 ± 56.65
TMP	19	14	1.08 ± 0.14	491.95 ± 25.76	580.37 ± 29.79	542.16 ± 26.78	24.37 ± 3.56	141.00 ± 17.28	67.26 ± 8.52	40.68 ± 5.73	88.42 ± 16.43
GRW	39	14	7.48 ± 0.33	16.64 ± 0.48	21.59 ± 0.61	19.90 ± 0.48	1.37 ± 0.14	1.32 ± 0.13	62.17 ± 7.05	46.49 ± 4.11	4.95 ± 0.46
RUM	18	10	7.38 ± 0.77	20.22 ± 0.73	26.59 ± 0.95	24.27 ± 0.83	1.75 ± 0.19	2.13 ± 0.29	79.44 ± 7.41	48.46 ± 4.22	6.48 ± 0.78
BRM2	8	5	4.59 ± 0.72	25.25 ± 4.01	43.63 ± 7.87	35.00 ± 5.62	5.63 ± 1.59	6.83 ± 3.64	93.47 ± 4.87	13.03 ± 12.44	18.38 ± 4.39
RRM2	4	3	2.60 ± 0.31	34.63 ± 3.69	50.63 ± 3.13	41.88 ± 3.69	4.06 ± 0.44	6.77 ± 3.08	93.94 ± 3.36	4.07 ± 1.64	16.00 ± 1.08
LRM2	10	7	2.74 ± 0.29	27.48 ± 2.60	44.97 ± 3.53	36.59 ± 2.51	4.89 ± 0.84	7.75 ± 1.48	95.00 ± 1.41	0.91 ± 0.58	17.49 ± 3.37

All are expressed as estimated mean $\pm$ SE. stDev is the standard deviation of frequency points along the F_0 contour within calls, not the standard deviation from the mean F_0 among multiple calls.

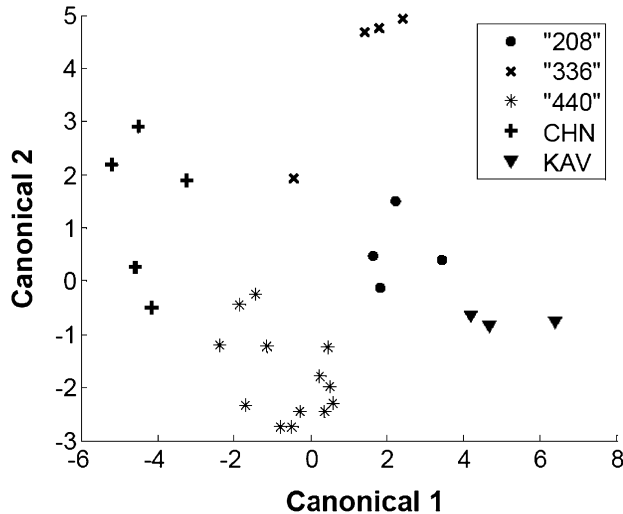


Figure 9. Growls by adult females are individually distinct. Individuals were maximally distinguishable based on the first two canonical dimensions (MANOVA, $df = 28$, $p_{\text{canon1}} \ll 0.0001$, $p_{\text{canon2}} = 0.0042$), which were linear combinations of the two harmonicity measures and all F_0 measures except 'range'.

Barks, roars and longroars

Barks were short, a-periodic calls that did not repeat. They were seemingly spontaneous vocalizations that accompanied group movement preceded by loud rumbling, or aggression accompanied by a swift lunge with physical contact such as pushing or biting. They were usually directed at other elephants, especially calves. Calves could target other animals such as birds, apparently in play.

Roars were given primarily during movement, but also during disturbance and distress. Longroars are longer vocalizations that occurred largely during separation, movement, searching and distress. Infrequently (6–7% of occurrences), this included calves who either sought out their mothers in order to nurse, or were interrupted while nursing. However, the caller was not necessarily isolated. On one occasion the caller (a sub-adult female) was surrounded by a herd of familiar individuals but persisted in calling until joined by an adult female who had been absent. Longroars accompanied behaviour indicating high arousal, such as lifted head, extended tail and brisk movement. Both roars and long roars could occur in bouts by a single individual, or as choruses by several individuals. Bouts of longroars by a single caller

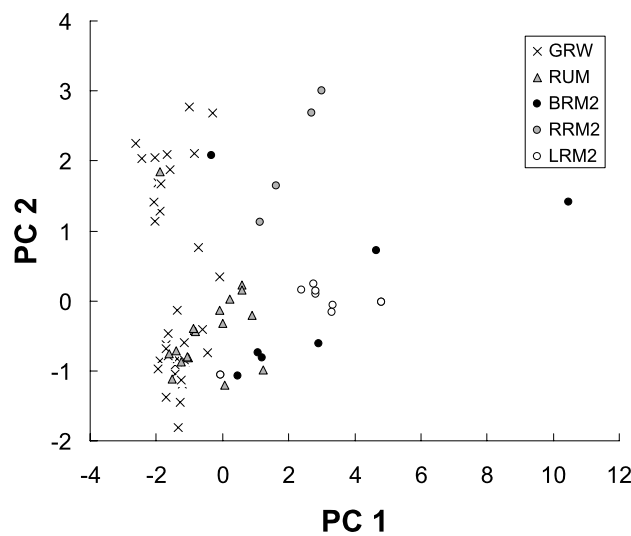


Figure 10. Principal components of growls, rumbles and second segment of combination calls based on F_0 contour. Principal component loadings are provided in table 4. Higher values of PC 1 correlate with higher F_0 values. Values of PC 2 correlate positively with percent maximum (highest frequency occurs towards end of call) and negatively with percent minimum (lowest frequency occurs towards beginning of call). The reverse is true of PC 1. Calls with falling F_0 , therefore, have negative values on PC 2 and positive values on PC 1. The second segment of combination calls (circles) are distinguished from growls and rumbles primarily along PC 1 but are more similar to rumbles than growls. PC 2 distinguishes these segments from one another.

Table 4. Loadings for first two principal components for growls, rumbles and the second segment of bark–rumbles, roar–rumbles and longroar–rumbles.

	PC 1	PC 2
Duration	−0.29577	−0.37372
Min	0.35849	0.18488
Max	0.43359	0.056302
Mean	0.427	0.088993
stDev	0.38734	−0.0394
Abs slope	0.3892	0.10737
% min	0.16356	−0.74984
% max	−0.29227	0.48962
Eigenvalue	5.10	1.27
% of variation captured	63.72	15.87

Only the first two principal components had eigenvalues > 1.

could at times last an hour or more. Calling could also occur from multiple locations separated by a kilometer or more, out of view of observers. Roars and longroars were audible to us from such distances even under windy conditions.

Combination calls

Croak-rumbles occurred only in bouts that contained rumbles, typically in choruses where at least two or more individuals called in rapid succession. Chirp-rumbles were observed only on three occasions, and occurred in bouts of 3 or more calls by adult females who were among companions. On two of these occasions the caller was then approached by a juvenile or infant calf but on the third occasion vegetation obscured our view of any small calves.

Bark-rumbles, roar-rumbles and longroar-rumbles primarily occurred during movement and searching. All three calls may be functionally similar graded vocalizations though a small number of bark-rumbles (2.3%) and roar-rumbles (6.3%) also were produced by calves wanting to nurse.

Squeaks, squeals, trumpets, trunk bounces and blows

Over 90% of squeaks (termed 'chirps' by McKay et al., 1973 and Nair et al., 2009) occurred in bouts of three or more in rapid succession (<1 s apart) in response to disturbance, and were usually accompanied by blows (loud, rapid exhalations), trunk-bounces, where the trunk is curled beneath the chin then rapidly extended so that the leading edge hits the ground loudly (Figure 7c, termed 'boom' by McKay, 1973), accompanied by behaviour indicating fear or excitement. One recording session contained single calls separated by 30–60 s or more, however the caller and context were unidentified. Other vocal activity at such times included squeals. They were sometimes produced by groups of females and sub-adults when approached by a male in musth, but were more often directed at humans.

Trumpets always occurred in response to disturbance, and sometimes were accompanied by squeaks. They could be accompanied other threats (e.g., throwing objects, charging).

Courtship and musth chirp-rumble

In, 2007, 2008 and 2009 calls were observed from three separate males in musth that appeared distinct from all other vocalizations (Figure 11). We



Figure 11. Musth chirp-rumble.

have not observed such vocalizations from female Asian elephants, although it is possible they produce them. We have also not observed calls corresponding specifically to mating or estrous, while such calls have been documented in African elephants (Poole et al., 1988; Langbauer, 2000). Moreover, even though musth males were observed several times per month, this vocalization was rarely observed. The first component was a brief (0.25 s), softly audible segment at approx. 250 Hz, followed by a longer low-frequency segment (0.75 s) at approx. 60 Hz. In some cases the second segment was absent altogether. Calls separated by 11–25 s occurred in bouts which could last several hours continuously. It is possible that the first segment is the consequence of inhalation rather than exhalation. On two out of the three occasions the calling male was engaged in a protracted contest with another male who was also in musth. On the third occasion the male was calling while courting and guarding a female in oestrus.

Successful matings rarely occurred within clear view of observers, and were not accompanied by many vocalizations. On occasions when such behaviour was accompanied by sound, the caller could not be confirmed but we suspected the female who was mounted. Only single vocalizations such as a trumpet or roar were heard.

Discussion

The Asian elephant acoustic repertoire consists of at least 14 different types of vocalizations in addition to one non-vocal acoustic signal (the trunk-

bounce) which seem to serve predominantly social and spatial functions. Of these vocalizations, nine have been previously described by McKay (1973) and Nair et al. (2009). 'Trumpets' are characteristic proboscidean vocalizations that are similar for both Asian and African species, but are confusingly also described as 'a long high-amplitude squeak' by McKay (1973). 'Growl' is the same vocalization described by McKay (1973). It is possible that they are acoustically similar to some subset of the vocalizations produced by African elephants which are uniformly termed 'rumbles' (Berg, 1983; Langbauer, 2000; Soltis et al., 2005b). 'Rumble' is comparable to low-frequency vocalizations of African elephants (Berg, 1983; Nair et al., 2009), possibly described by McKay (1973) as 'motorcycle'. 'Squeak' was previously termed 'chirp' by the same authors, but McKay (1973) and Langbauer (2000) describe it also as 'multiple short squeaks'. I prefer 'squeak' to refer to the single call, which is intuitive and simple. Multiple squeaks in succession merely constitute a bout, as with any other vocalization. 'Squeals' do not appear to have been described. Both squeaks and squeals appear unique to Asian elephants (see also Nair et al., 2009). 'Roars' have previously been seen as a single category (McKay, 1973). While I split them into two categories based on duration, they may differ little functionally. 'Barks', which I define by their short duration, have never been described before, though McKay (1973) mentions the ambiguous 'snort'. None of the 'combination' calls have been recognized as such previously, though 'bark-rumble' may be comparable to the 'Rev-followed-followed-by-rumble' of *Loxodonta* reported from captive study (Leong et al., 2003b) and roar-rumbles and longroar-rumbles may be acoustically similar to calls produced by *Loxodonta* calves when nursing, which adults are not reported to produce (personal observation). It is not clear whether *Loxodonta* produce 'longroars'. 'Musth chirp-rumbles' had neither been documented in the wild, nor, to my knowledge, had they been reported in captivity for *Elephas*. They appear acoustically different from the 'musth-rumble' of *L. Africana* (Poole et al., 1988), and may be similar to contest vocalizations given in species such as chacma baboons (Kitchen et al., 2003).

It is unclear whether harmonics in trumpets are caused by source or filter effects. Double-voicing, where two sets of closely-spaced harmonic bands appear within a single vocalization, is evident in trumpets (Figure 2d). Birds such as king penguins use this feature, which originates in the uniquely avian syrinx, to identify each other (Aubin et al., 2000; Lengagne et al., 2001). It

is not clear what the physiological cause of this could be for elephants, aside from perhaps the dual nasal passages of the trunk, or whether it serves any function.

Growls are individually-distinct and, therefore, may be used for locating one another or maintaining spacing among members of different social groups. It is likely that other calls are also individually-distinct, though I could not test this. As sample sizes for some individuals in this study were small, it may be worth repeating analyses with additional data and call types. Figure 10 also seems to suggest that growls fall into two groups along the second principal component. Because the second principal component is determined by the location of the minima and maxima of calls, those that show little change in the fundamental frequency can by chance show minima at either the beginning or end and, thus, appear different along these measures. Rumbles of African elephants have been shown to vary contextually (Wood et al., 2005; Leighty et al., 2008). Growls as well as rumbles may in fact be differentiable further using other acoustic and behavioural measures; however, large sample sizes are needed.

Rumbles and growls comprise the bulk of vocalizations observed. Although the use of low-frequency communication by African elephants to coordinate movement over long distances has been much discussed (Payne et al., 1986; Langbauer et al., 1991; McComb et al., 2003; Garstang, 2004), the function of similar calls by Asian elephants has not been explored. Rumbles and growls may permit coordinated activities in dense vegetation, however combination calls and longroars also occur with search behaviours. The significance of call types that appear functionally redundant deserves exploration. It is possible that growls are simply softer, less powerful, calls which carry less than 100 m whereas rumbles enable coordinated activity over greater distance and longroars or combination calls are used to make contact still further away, as first proposed by McKay (1973). This proposition was made prior to the discovery of infrasonic communication. It is also possible that these calls reflect different levels of excitement. These hypotheses remain to be tested. Though we did not assess the power level of these calls directly, future study of their transmission properties or playback experiments would provide a better understanding of the spatial extent of these calls, and their interplay with social dynamics.

Combination calls are composed of distinct segments. Segments appear acoustically similar to calls that occur singly. The leading segments of all

combination calls grade by duration. Differences in durations of the first segment of 'longroar-rumbles' and 'longroars' likely arise because they reach a physiological limit, lung capacity; hence, the first segment of longroar-rumbles are not the same length as longroars. The second segments of combination calls overlap in acoustic space with rumbles more than with growls. Principal component loadings suggest that acoustic differences between these segments are largely due to the fact that these segments show falling F_0 contours as a result of being connected to a segment with higher frequencies at onset, whereas rumbles and growls show level F_0 contours. I maintain the term 'rumble' to describe the second segment of combination calls for parsimony.

There are suggestions of age and sex differences in call production, such that repertoires may reflect the divergent life histories of males and females (see also Nair et al., 2009). Younger individuals only appear to produce a subset of the calls produced by adults and adults rarely produce certain vocalizations. It is not clear whether this is due to developmental reasons, or because certain contexts are more relevant to certain age and sex classes. For instance, adults may not panic when separated from companions and, thus, rarely produce longroars, though able to do so. Sub-adults and juveniles, as they wean and become independent, are also simply more likely to get lost. Likewise, adult males may not produce calls functioning in social cohesion if they do not maintain such relationships.

Vocalizations produced by infants may sometimes be superficially different to those of adults (e.g., higher frequency) simply due to differences in vocal physiology rather due to later modification through learning. However, some species do show changes in their vocal repertoires with age (Elowson et al., 1998; Tchernichovski et al., 2001). In certain non-human primates, calls may be acoustically well-formed from infancy but the appropriate contexts for production have to be learned through experience (Seyfarth & Cheney, 1986). While only passerines, cetaceans and humans are commonly acknowledged to be capable of vocal imitation in addition to contextual learning (Janik & Slater, 1997; Watwood et al., 2004; Pepperberg, 2006), there is some evidence that African elephants in captivity may learn atypical vocalizations (Poole et al., 2005). We have recorded sub-adults, and occasionally adult females in the wild producing calls which defy classification, which are not included here. It is an open question whether Asian elephants demonstrate vocal learning or convergence among individuals. Indeed, frequency matching seems common when individuals chorus (personal observation).

It is intriguing that out of the 258 measurable calls described by Nair et al. (2009), close to 30% are trumpets, 22% are roars, 26% are 'chirps' (squeaks) and 22% are rumbles. This call type distribution is quite different from those reported here, and it is not clear whether other call types were observed at all and whether this distribution is representative of all observed vocalizations or merely those suitable for acoustic analysis. Three out of four of these call types occur primarily during disturbance whereas the fourth can also occur during disturbance, which is reinforced by the contextual descriptions by Nair et al. (2009). If such differences are not due to our methods of study or call classification, it suggests the Uda Walawe elephants are less disturbed, possibly due to differences in levels of habituation of the two study populations. On the other hand, as Asian elephants have evolved into populations that have been isolated from one another by the sea as well as human activities, acoustic signals could have differentiated sufficiently enough to constitute dialects. Further study of the development and function of vocal communication in different populations of Asian elephants is likely to broaden and inform the description set forth here for comparison with African elephants and other species.

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